

=== Harwell RSX Characterization Report ===

Each resistor undergoes characterization at multiple temperature points to develop a precise model for calculating resistance across its operating range. Our modern resistors use a proprietary metal foil design that depending on the unique assembly of each resistor, we may characterize it via a polynomial or third-degree fit.

=== Instrument Summary ===

Date: 2025-08-21

Model: Harwell RS-7

Value: 1 Ω

S/N: RS71003

=== Calibration Data ===

Calibrated Resistance: 0.99968542 Ω at 23°C

Expanded Uncertainty of Measurement, k=2: 0.17 $\mu\Omega/\Omega$

Measurement Current: 100 mA

=== Temperature Characterization Data ===

α ($\Omega/^\circ\text{C}$): -8.48e-07

β ($\Omega/^\circ\text{C}^2$): 5.37e-08

Linear TCR (18-28°C): -0.80 PPM/ $^\circ\text{C}$

To calculate the resistor value at a temperature other than 23°C, use the following model where T is the desired temperature in $^\circ\text{C}$:

$$R(T) = R_{(23)} + \alpha(T - 23) + \beta(T - 23)^2$$

Note: This model may not be accurate outside the characterized temperature range.

Temperature ($^\circ\text{C}$)	Temperature ($^\circ\text{F}$)	Resistance (Ω)	Deviation (ppm)
18.0	64.4	0.99969100	5.58
18.5	65.3	0.99969032	4.91
19.0	66.2	0.99968967	4.25
19.5	67.1	0.99968905	3.63
20.0	68.0	0.99968845	3.03
20.5	68.9	0.99968788	2.46
21.0	69.8	0.99968733	1.91
21.5	70.7	0.99968681	1.39
22.0	71.6	0.99968632	0.90
22.5	72.5	0.99968586	0.44
23.0	73.4	0.99968542	0.00
23.5	74.3	0.99968501	-0.41
24.0	75.2	0.99968463	-0.79
24.5	76.1	0.99968427	-1.15
25.0	77.0	0.99968394	-1.48
25.5	77.9	0.99968364	-1.79
26.0	78.8	0.99968336	-2.06
26.5	79.7	0.99968311	-2.31
27.0	80.6	0.99968289	-2.53
27.5	81.5	0.99968269	-2.73
28.0	82.4	0.99968252	-2.90

Resistance vs Temperature

